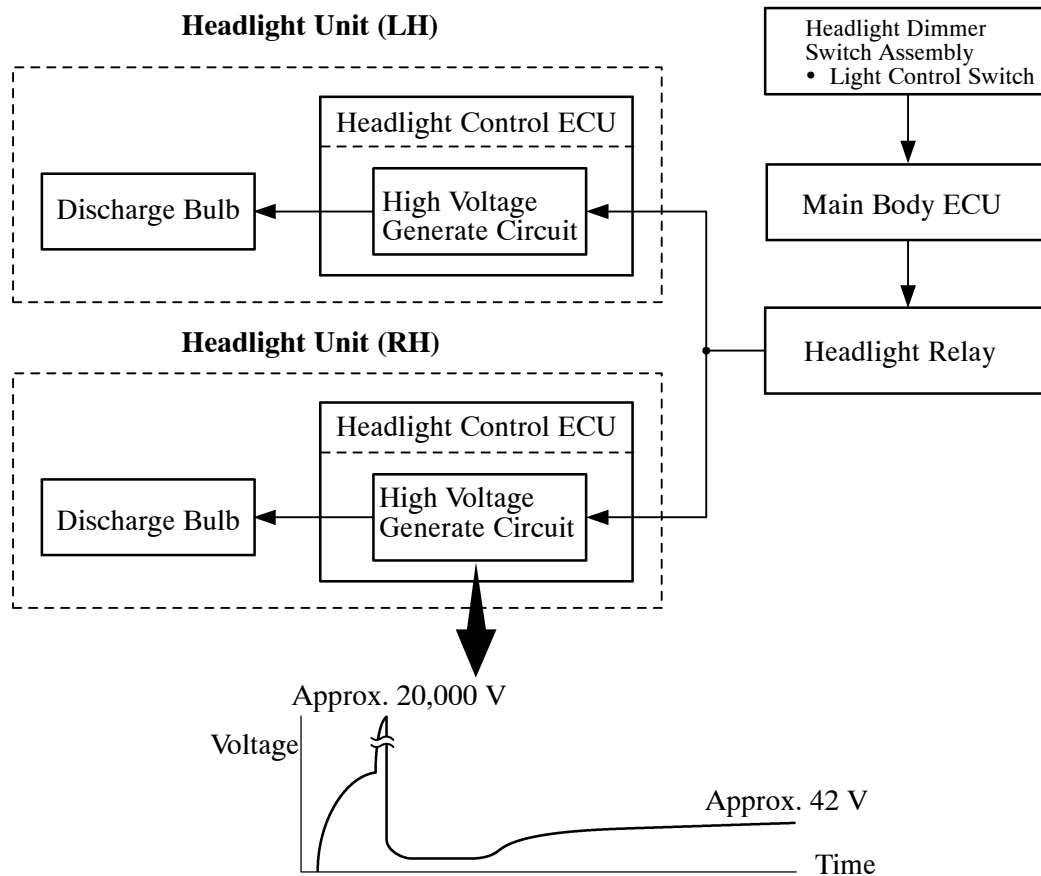


■ HID HEADLIGHT SYSTEM

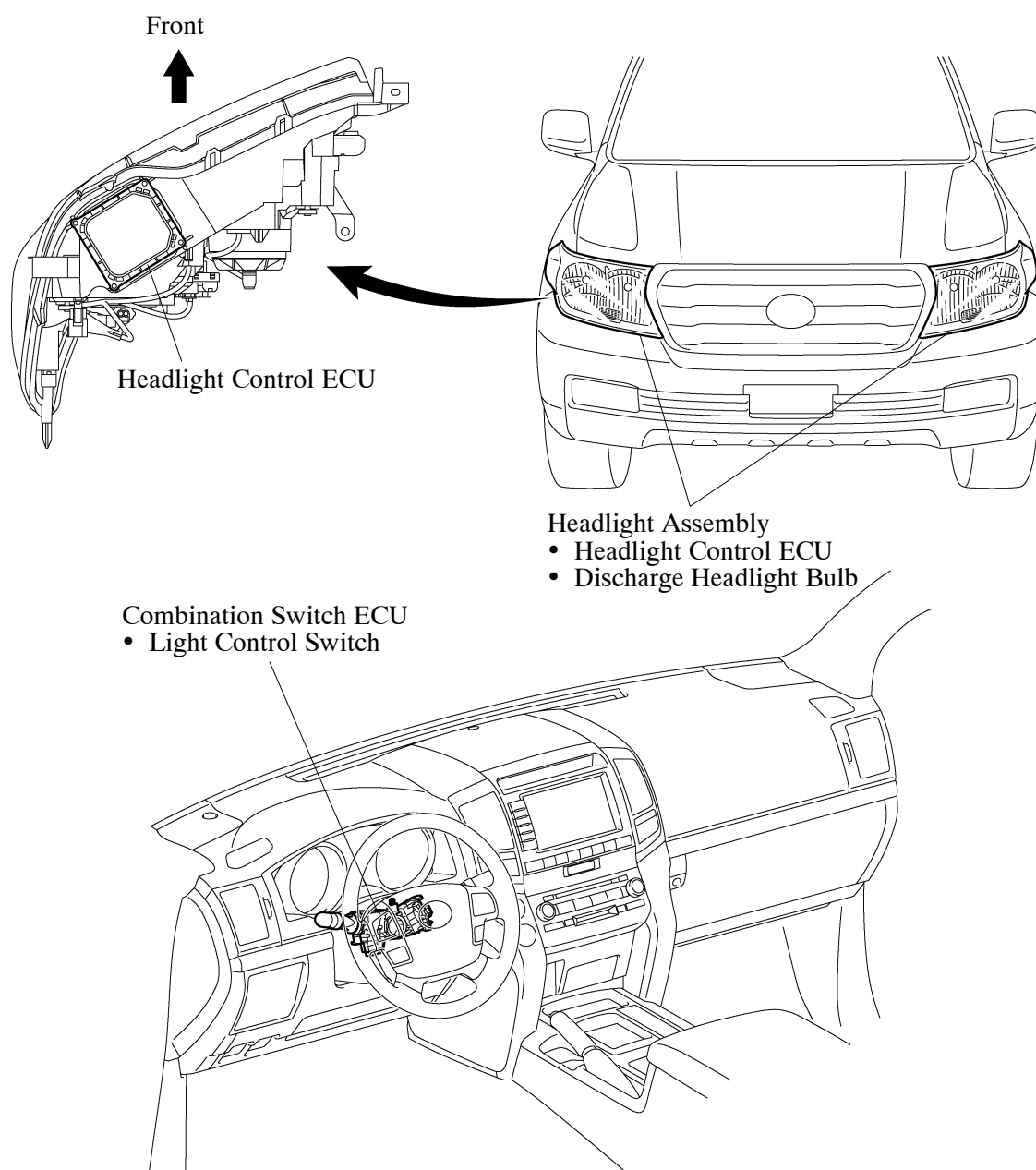
1. General

- The HID (High Intensity Discharge) headlight system uses a discharge bulb as its light source for the Hi beam. Discharge bulbs are superior to halogen bulbs.
- Discharge bulbs have the following features.
 - The light emitted by the bulb is close in color to sunlight. The light shines ahead over a broader area and further forward, increasing the area visible to the driver.
 - Less power is consumed.
- This system consists of discharge bulbs and headlight control ECUs.
- Headlight control ECU transforms the voltage that is input from the battery to a high voltage of up to 20,000V and applies it to the discharge bulbs in order to illuminate them.
- A fail-safe function is provided as a countermeasure against the high voltage that is generated in case that a problem occurs in the headlight system.

► System Diagram ◀



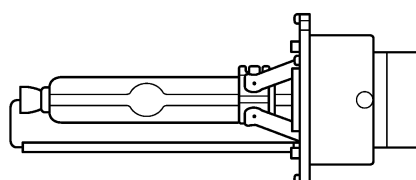
2. Layout of Main Components



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3. Discharge Bulb

Instead of the filament contained in an incandescent bulb, a discharge headlight bulb contains an arc tube, which is filled with xenon gas, and metal halide.



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4. Fail-safe Function

The light control ECU executes the fail-safe actions listed below in accordance with the problem that has been detected.

Problem	Outline
Detection of Abnormal Input Voltage	If the voltage that is input to headlight control ECU deviates from the normal operating voltage (9 - 16 volts), the headlight control ECU stops illuminating the headlights. It resumes illuminating the headlights once the voltage reverts to the operating voltage range. However, if the input voltage decreases after the headlights have illuminated, the headlights will remain illuminated until the input voltage is insufficient to light the bulbs.
Detection of Abnormal Output (Open Circuit or Short Circuit) or Flashing Bulb	If an abnormal condition (open or short) occurs in the voltage that is output by headlight control ECU, or if the bulb flashes, the headlight control ECU stops illuminating the headlights and will maintain this state until the power is reinstated. Power is reinstated by turning the headlight control switch from OFF to ON.
Detection of Bulb Open	If a bulb is not inserted in its socket, the headlight control ECU stops generating high voltage until the bulb is inserted correctly and the power is reinstated. Power is reinstated by turning the headlight control switch from OFF to ON or turning the engine switch from OFF to ON.

5. Precautions for HID Headlight System

- When the HID headlights illuminate, a high voltage (approximately 20,000V) is applied momentarily to the bulb sockets, which could lead to a serious accident. Never connect the tester to the high voltage socket of HID headlight for measurement, as this may lead to a serious accident because of high voltage.
- Whenever inspecting the HID headlight system, make sure water or rain is not present in order to prevent electric shock, the light control switch is OFF, the battery terminal is removed, and the connector of the headlight control ECU is disconnected.
- Whenever operating the HID headlight, make sure it is only after assembly has been completed and never operate without bulbs installed.
- Do not operate the HID headlight using a power source other than the vehicle's.
- When there is a defect in the HID headlight or any shock has been applied to it, replace the bulb with a new one.
- A discharge headlight bulb reaches a high temperature when it is illuminated. For this reason, the life of the bulb could be shortened if any oil comes in contact with the glass portion of the bulb. Do not touch the glass portion of a bulb with bare hands.